

Three Ways to Miss a Grasp: A Same-Metric Comparison of Rule-Based, Learned, and Zero-Shot Vision-Language Grasp Prediction

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Code and data: <https://github.com/anishtalla27/Vision-Based-Grasping>

Abstract—Rule-based, learned, and vision-language approaches to robotic grasp prediction are usually evaluated under different inputs, output formats, and scoring rules, which makes their published accuracies difficult to compare. This paper provides a controlled evaluation of the three paradigms. A detector-assisted heuristic, a fine-tuned convolutional network, and zero-shot GPT-4o are tested on one object-wise split of the Cornell Grasping Dataset, return the same grasp rectangle, and are scored by a single implementation of the standard rectangle metric. Uncertainty is estimated by resampling whole objects, because the test images are repeated views of only 35 objects. The learned predictor is the most accurate, the heuristic follows, and the language model is far behind both. A shared failure taxonomy shows that each system breaks at a different stage of the task. A follow-up experiment then tests the most natural explanation for the language model’s result, that it identifies a suitable grasp but cannot write it as pixel coordinates. When the same model only has to select one of several numbered grasps drawn on the image, its accuracy remains at the level of random selection. The limitation lies in choosing the grasp, not in expressing it.

Index Terms—robotic grasping, vision-language models, GPT-4o, Set-of-Mark prompting, Cornell Grasping Dataset, clustered bootstrap

1. Introduction

Before a parallel-jaw gripper closes, a robot must settle three geometric quantities: where the fingers land, how the gripper is rotated, and how far the jaws open. That decision can be written by hand as a set of rules, learned by a neural network from labeled examples, or requested from a general-purpose vision-language model (VLM) that was never trained for grasping. The three options differ greatly in setup cost. A rule needs no labels, a network needs a labeled dataset and training, and a VLM needs only a prompt. Their reported accuracies, however, are rarely comparable, because the input, the output format, or the scoring rule changes from one method to the next.

This paper treats the comparison as a measurement problem. One system of each kind is built, and everything

around the systems is held fixed. Each system receives the same 123 held-out RGB images, returns the same rectangle representation, and is scored against the same Cornell labels by one implementation of the standard rectangle test [1], [2]. Object identities, which the dataset does not provide, are reconstructed so that no physical object appears in both training and test data. Because several test images show the same object from different angles, all statistics are computed over objects, not images. The VLM, whose output varies between calls on identical input, is queried five times per image so that this variation can be measured.

The contributions are as follows.

- A same-metric evaluation of a rule-based, a learned, and a zero-shot VLM grasp predictor on a leak-free object-wise split, with object-clustered confidence intervals and paired significance tests.
- A shared failure taxonomy, together with a spatial check on the predicted grasp center, that locates the stage at which each system fails.
- A controlled follow-up experiment (System D) that removes the VLM’s coordinate output and tests whether its failure is one of expression or of grasp selection.
- A label-free geometric baseline that emerges from that experiment and matches the detector-based heuristic.

The goal is not to exceed the state of the art. Dense grasp detectors already report higher Cornell scores [4]. The goal is to compare three practical starting points under a test that does not change with the method.

2. Related Work

Jiang *et al.* introduced the Cornell dataset and its rectangle representation of a grasp, which stores the center, orientation, opening, and jaw depth [1]. Under the usual Cornell test, a prediction passes when it lies within 30° of a labeled grasp and its intersection over union (IoU) with that grasp exceeds 25% [2]. Redmon and Angelova regressed a single rectangle with a convolutional network and reported 84.9% object-wise accuracy using RGB-D input [3]. GR-ConvNet and other pixel-wise models report higher scores,

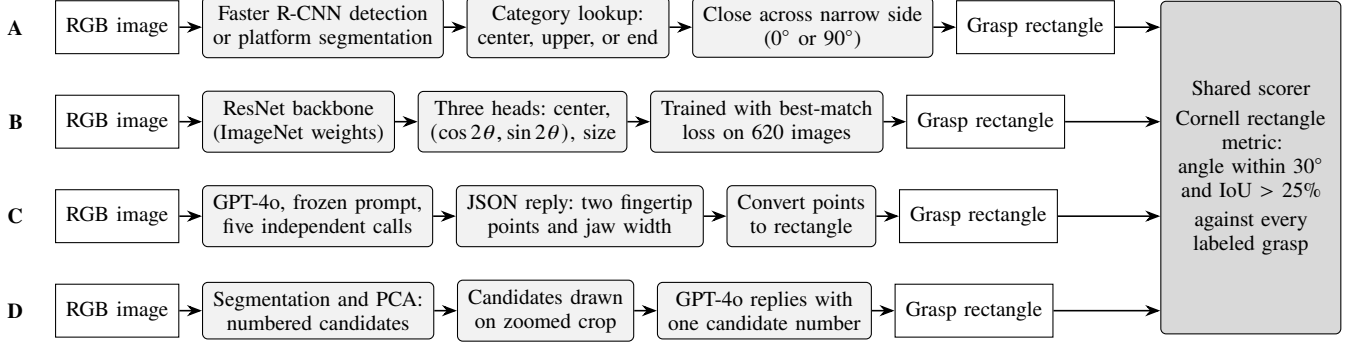


Figure 1. The four systems. Each receives the same test image and produces a grasp rectangle in the native 640×480 frame, and all rectangles pass through one implementation of the Cornell metric. Systems C and D use the same language model and differ only in what the model is asked to output.

although they produce a dense prediction over the image and do not regress one rectangle [4].

VLM-based grasping systems generally avoid asking the language model for final pixel coordinates. Set-of-Mark prompting has the model choose among regions already marked on an image [5]. VLAD-Grasp has a VLM draw the grasp and then recovers a pose through depth prediction, segmentation, and point-cloud alignment [6]. System C in this paper tests the simplest possible pipeline, in which GPT-4o is shown the image and asked for the coordinates directly. System D then moves toward the Set-of-Mark setting. The scores of both systems describe those two ways of querying the model and do not bound what VLM-based grasping can achieve.

3. Experimental Design

Figure 1 summarizes the four systems. This section describes the protocol and each system. All outcomes, including those from development-stage checks, are reported in Section 4.

3.1. Dataset, Split, and Metric

The Cornell Grasping Dataset contains 885 RGB-D images of household objects, most with several labeled positive grasp rectangles. Only the RGB channels are used here. Two images were removed because their object boundaries could not be resolved, which leaves 883.

The dataset does not include object identifiers, so they were reconstructed from the frame sequence. The object was segmented from the photography platform, and consecutive masks were compared using color and area descriptors that tolerate rotation. The two possible errors are not equally harmful. If two objects are merged, the combined group still stays within one partition. If one object is divided in two, its images can leak into both training and test data. The decision thresholds were therefore set to favor merging. An automated first pass assigned a decision and a confidence level to each ambiguous boundary, and a blind audit of 30 boundaries measured its agreement with manual labels (Section 4.1). Every uncertain decision and every high-risk

“different object” decision was then resolved manually, and a final check confirmed that no object appears in more than one partition.

Each prediction is scored against every positive rectangle on its image, and a match to any of them counts as correct. Before the evaluator was used at scale, it was verified against hand-computed cases, rectangle round trips, and an independent rasterized IoU calculation. All predictions are converted to the native 640×480 coordinate frame before scoring.

3.2. System A: Detector-Assisted Heuristic

System A begins with a COCO-pretrained Faster R-CNN with a ResNet50-FPN-v2 backbone [7]. A fixed lookup table maps the detected category to a center, upper, or end grasp, and the gripper closes across the narrow dimension of the bounding box. The global opening and jaw-size constants were calibrated on training labels only. Because the COCO vocabulary covers few Cornell objects, a platform-segmentation fallback supplies a box when the detector returns nothing. By construction, the system can output only 0° or 90° .

The system was revised once after its first test evaluation. Rendered predictions showed that some accepted detections covered background clutter instead of the object on the platform. A guard was added that requires the detection to contain the centroid of the segmented object, and the global constants were recalibrated on training data. Because test output prompted this change, the corrected score is treated as post-hoc throughout, and the pre-correction score is reported alongside it.

3.3. System B: Learned Predictor

System B replaces the ImageNet classifier of a ResNet [8] with separate heads for center, orientation, and size. The orientation head predicts $(\cos 2\theta, \sin 2\theta)$, since a parallel-jaw grasp is unchanged by a 180° rotation. For images with several labels, the loss is computed against every rectangle and only the best match is backpropagated. Averaging over

all valid labels could place the target between two good grasps, at a location where no valid grasp exists.

Training proceeded in two rounds on the 620 training images, with affine augmentation that moves the rectangle corners together with the pixels. In the first round, a small custom CNN, ResNet18, and ResNet34 were each trained once with a constant learning rate. The architecture, checkpoint, size-loss weight, and the custom CNN’s learning rate were selected on validation data, and the selected model was frozen before the test split was opened.

The first round relied on a single seed and on the best of roughly one hundred noisy validation epochs. The second round kept the model, loss, data, and split, and changed only the training recipe: cosine learning-rate decay over a fixed 100 epochs, an exponential moving average of the weights, three seeds per architecture, and no checkpoint selection. Test-time augmentation over eight dihedral views, combined by taking the medoid rectangle, was adopted on the basis of validation accuracy alone. Four selection rules were written down before the test split was opened a second time, and every configuration that was evaluated on test data is reported.

3.4. System C: Zero-Shot VLM

System C sends the native RGB image to GPT-4o through OpenRouter [9], with no grasp-specific training. The prompt requests two fingertip coordinates and a jaw width in JSON, which together define the rectangle. The prompt was developed on 30 training images and frozen before testing.

Each test image was submitted in five independent single-turn calls at the provider’s default temperature. Transport failures were retried. Replies that arrived but failed to parse were not, because retrying them would convert a single-call baseline into a best-of-several system. The primary score pools the five exchangeable runs to estimate single-call accuracy. Best-of-five accuracy is reported only as an upper bound, since it requires five calls and an oracle that knows which call is correct.

3.5. System D: Marked-Candidate Selection

If GPT-4o can identify where to grasp but cannot express the location as coordinates, removing the coordinate output should recover its choice. System D shows the same model the same test images with candidate grasps drawn on a zoomed crop (Fig. 2) and asks for the number of the candidate it would use. Candidates are generated from the platform-segmentation mask alone. Principal-component analysis gives the centroid, long axis, and extent of the object, and candidates are placed at positions along the long axis with four jaw-travel directions 45° apart, so that every possible grasp angle lies within the metric’s tolerance of some candidate. Opening and jaw width reuse System A’s training-calibrated constants, and the numbering is reshuffled on every call. Because the menu is fixed before the model sees it, each run has a known floor (the expected



Figure 2. A System D input from the twelve-mark menu. Each numbered mark is a candidate grasp, with pads at the two fingertip positions. On all five repeats the model chose mark 6, the pinch along the length of the object. Mark 10, which crosses the object at its center, passes the metric.

accuracy of a uniformly random choice) and a known ceiling (the share of images on which any candidate passes).

Three menus were evaluated, each on the full test split with five repeats. The first has twelve marks at three positions. The second has four marks at the centroid only, drawn at real size. The third has the same four marks drawn at one uniform length, with a prompt stating that length is not to scale. The last two menus control for a confound: on thin objects the correct marks are the smallest ones on the image, so a preference for large marks would be indistinguishable from a preference for a particular grasp direction. The prompt was developed on 30 training images for parse rate and glyph legibility only, and each menu was run on the test split exactly once.

3.6. Statistical and Failure Analysis

The 123 test images are repeated views of 35 unseen objects and are not independent samples. All 95% confidence intervals therefore come from 20,000 object-clustered bootstrap resamples, in which every view of an object and all five VLM calls on each view are resampled together. Paired differences between systems use the same resamples, and significance is assessed with 20,000 object-level label-swap permutations.

For the failure analysis, each attempt receives exactly one label: correct, angle criterion failed, overlap criterion failed, both failed, or no prediction. A second check records whether the predicted center falls inside the convex hull of all labeled grasp corners on the image. This is a loose spatial test of whether a prediction is near any annotated grasp, and it makes no claim about physical grasp success.

TABLE 1. OBJECT-WISE CORNELL SPLIT.

Partition	Objects	Images
Train	164	620
Validation	35	140
Test	35	123
Total	234	883

TABLE 2. TEST ACCURACY WITH OBJECT-CLUSTERED 95% CONFIDENCE INTERVALS. “OBJ.-WT.” GIVES EACH OF THE 35 TEST OBJECTS EQUAL WEIGHT.

System	Acc.	95% CI	Obj.-wt.
A: heuristic, pre-correction	40.7	–	–
A: heuristic, post-hoc fix	57.7	[46.2, 68.2]	51.3
B: ResNet18, round 1	79.7	[70.8, 87.1]	75.4
B: ResNet34, round 2, 3 seeds	84.0	[73.9, 92.1]	–
C: GPT-4o, single call	12.4	[8.0, 17.4]	10.5
C: GPT-4o, best of five	35.0	[24.6, 45.5]	–

4. Results

4.1. Data Split and Boundary Audit

Table 1 gives the final split, generated with seed 42. In the blind 30-boundary audit, the automated first pass agreed with manual labels on 86.7% of the boundaries it marked confident and on 40.0% of those it marked uncertain. The low agreement on uncertain cases is the reason all of them were decided manually.

4.2. Overall Accuracy

Table 2 and Fig. 3 report test accuracy for every system, and Table 3 reports the paired differences. The ordering is unambiguous. The learned predictor leads, the heuristic follows, and single-call GPT-4o is far behind both, including the heuristic’s pre-correction score. Every paired difference between the three paradigms excludes zero, and this holds even when GPT-4o is credited with its best-of-five upper bound. Weighting each object equally instead of each image (last column of Table 2) changes the size of the gaps but not their order. Clustering by object widens every interval relative to an image-level analysis, which shows how much an image-level analysis would overstate the precision of a 123-image test set.

4.3. Stability of the Learned Predictor

Table 4 places the two training rounds side by side. The second round does not show a statistically reliable improvement over the first (Table 3). Its value is that it measures the uncertainty the first round left unmeasured. Rerunning the first-round recipe at new seeds produced higher scores than the original run, which indicates that 79.7% was a low draw from that recipe. The first round also suggested that ResNet18 outperformed ResNet34 by nine points, but under the stable recipe the two architectures are level on validation

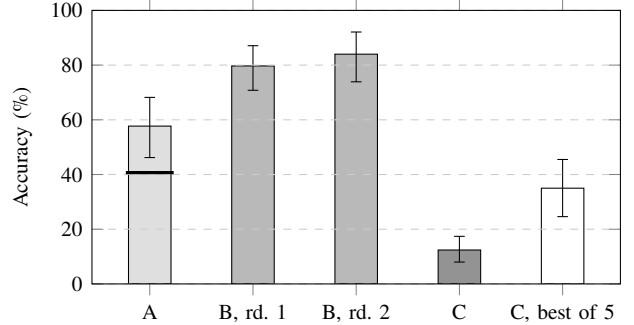


Figure 3. Test accuracy with object-clustered 95% intervals. The horizontal rule on bar A marks the heuristic’s pre-correction score. The white bar is an upper bound that requires five calls and an oracle.

TABLE 3. PAIRED ACCURACY DIFFERENCES IN PERCENTAGE POINTS, WITH OBJECT-CLUSTERED BOOTSTRAP INTERVALS AND PERMUTATION p -VALUES.

Comparison	Diff.	95% CI	p
B round 1 – A	22.0	[9.7, 35.1]	0.0031
B round 2 – A	26.3	[14.8, 38.3]	0.0002
B round 2 – B round 1	4.3	[−5.8, 13.7]	n.s.
A – C, single call	45.4	[33.8, 55.6]	<0.0001
A – C, best of five	22.8	[9.2, 35.6]	0.0060

data (81.2% and 81.4%) and within 1.4 points on test data. That apparent architecture effect was run-to-run variation, and a single-seed comparison would have reported it as a finding.

4.4. Failure Patterns

Table 5 separates the systems more clearly than accuracy does, because each system loses most of its attempts in a different row. System A’s angle-only failures are consistent with a rule restricted to two orientations: the rectangle overlaps a labeled grasp, but the object lies diagonally. System B rarely misses on angle alone, and most of its failures are overlap-only, so placement and sizing are its remaining weakness. System C fails both criteria on the majority of its calls, which indicates that its rectangles are usually not near a labeled grasp at all.

The center-hull check in Table 6 supports that reading. Systems A and B place almost all of their centers within the region spanned by the labeled grasps, while System C places more than half of its centers outside it. The three rates are not a measure of the same skill. System A starts from a detected or segmented object, and System B was trained on Cornell images, whereas System C may place its coordinates anywhere in the frame. The check is used only to locate the stage at which System C’s failures begin. Figure 4 shows the behavior on one test image.

System C is also inconsistent with itself (lower half of Table 6). On roughly a third of the test images, identical pixels produced a passing grasp on some calls and a failing grasp on others. A single-call score therefore understates the spread a user would observe, and the best-of-five figure

TABLE 4. SYSTEM B ON THE 123 TEST IMAGES. ROUND 1 REPORTS THE VALIDATION-SELECTED CHECKPOINT. ROUND 2 REPORTS FINAL AVERAGED WEIGHTS WITH TEST-TIME AUGMENTATION, AS A MEAN OVER THREE SEEDS.

Round	Configuration	Test acc. (%)
1	ResNet18, seed 42	79.7
1	ResNet34, seed 42	70.7
1	Custom CNN, seed 42	23.6
1 recipe	ResNet18, seeds 0 and 1	82.9, 80.5
2	ResNet18 (83.7, 80.5, 83.7)	82.6
2	ResNet34 (84.6, 83.7, 83.7)	84.0

TABLE 5. SHARED FAILURE TAXONOMY. ENTRIES ARE COUNTS, WITH PERCENTAGES OF ALL ATTEMPTS IN PARENTHESES. ROUND 1 MODEL FOR SYSTEM B.

Outcome	A ($n = 123$)	B ($n = 123$)	C ($n = 615$)
Correct	71 (57.7)	98 (79.7)	76 (12.4)
Angle failed only	13 (10.6)	4 (3.3)	85 (13.8)
Overlap failed only	18 (14.6)	15 (12.2)	88 (14.3)
Both failed	13 (10.6)	6 (4.9)	365 (59.3)
No prediction	8 (6.5)	0 (0.0)	1 (0.2)

cannot be realized without an external signal that identifies the correct call.

4.5. System D: Selection Among Marked Candidates

Table 7 reports the three menus. On none of them does the model’s accuracy differ from random selection among the same candidates, although a passing candidate was available on most images. Removing the coordinate output therefore did not recover a correct choice. All 560 submitted replies parsed, so the result does not reflect a formatting problem.

The model’s choices are nevertheless systematic. On every menu it selected the long-axis direction well above the chance rate, and the preference persisted when all marks were drawn at the same length, which rules out mark size as the explanation. On the twelve-mark menu, a single candidate type, the pinch along the long axis near one end of the object, received 61% of all choices although it passes on only 17% of images. The short-axis candidate at the centroid, which passes on 55% of images, was chosen in 17% of calls on the uniform menu. The model’s written justifications described the chosen grasp in plausible terms, such as gripping “the narrow top edge near the center of mass”, and its stated confidence was “high” on 556 of 560 calls, of which 21% were correct.

The experiment also produced an unplanned baseline. Always selecting the centroid candidate that closes across the short axis, with no model involved, scores 55.3% (Table 7). This is level with System A and requires neither a detector nor a category table.

TABLE 6. PREDICTED CENTERS OUTSIDE THE CONVEX HULL OF ALL LABELED GRASP CORNERS, AND AGREEMENT AMONG SYSTEM C’S FIVE CALLS PER IMAGE.

Quantity	Value
A: centers outside hull	8 of 115 (7.0%)
B: centers outside hull	8 of 123 (6.5%)
C: centers outside hull	328 of 614 (53.4%)
C: range of the five runs	9.8% to 14.6%
C: mean pairwise rectangle agreement	22.0%
C: mean pairwise IoU	0.13
C: images where all five calls fail	80 of 123
C: images where all five calls pass	1 of 123
C: images with mixed outcomes	42 of 123

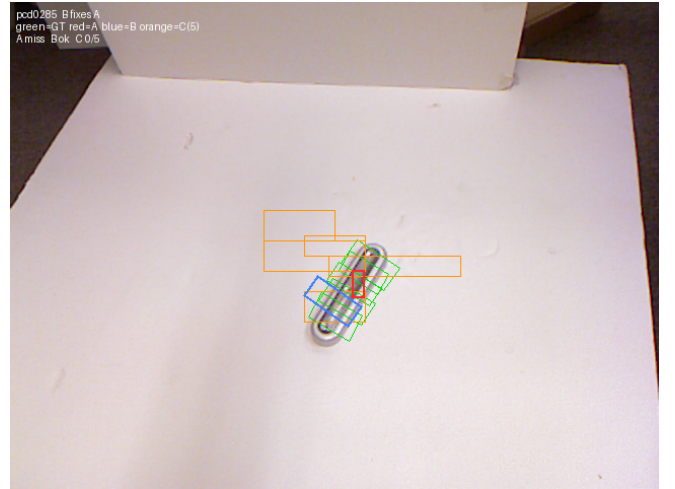


Figure 4. One test image in the shared coordinate frame. Labeled grasps are green, System A is red, System B is blue, and System C’s five calls are orange.

5. Discussion

5.1. Interpretation

The cost of each approach and its accuracy were ordered in the same way on this benchmark. The learned predictor required labels and training and was the most accurate. The heuristic required no labels and reached a moderate score, limited mainly by its two fixed orientations. Zero-shot GPT-4o required only a prompt and did not produce usable rectangles. That last result applies to direct coordinate prompting on this setup and should not be generalized to VLM grasping as a whole. Set-of-Mark and VLAD-Grasp both avoid asking the model for final coordinates, and both add components that System C lacks [5], [6].

The System C results initially suggested a coordinate-binding hypothesis. In a sample of its failures, the written rationale named a plausible part of the object while the coordinates landed elsewhere, and more than half of its centers fell outside the labeled region. Under that hypothesis, the model recognizes a suitable region in language and loses it when mapping the answer to pixels. System D tested the hypothesis directly, and the hypothesis was not supported.

TABLE 7. SYSTEM D ON THREE CANDIDATE MENUS, FIVE REPEATS EACH. “DIFF.” IS ACCURACY MINUS THE RANDOM FLOOR WITH ITS OBJECT-CLUSTERED 95% INTERVAL. “LONG AXIS” IS THE SHARE OF CALLS CHOOSING A GRASP WHOSE JAWS TRAVEL ALONG THE OBJECT’S LONG AXIS (CHANCE IS 25%).

Menu	Acc.	Floor	Ceil.	Diff. [95% CI]	Long axis
12 marks	18.7	20.1	79.7	-1.4 [-6.6, 4.0]	70%
4 marks, real size	26.5	24.6	65.9	+1.9 [-2.2, 6.3]	41%
4 marks, uniform	22.3	24.6	65.9	-2.3 [-6.3, 1.7]	45%
Geometric rule alone (centroid, short axis): 55.3 [41.8, 68.0]					

With coordinates removed and a passing candidate available on most images, the model still chose at the level of chance, and it kept its preference for closing along the long axis on menus with no end-straddling candidates and with uniform mark sizes.

A geometric reading of that preference is available. In a top-down photograph, two fingertip pads placed across the end of an elongated object appear to pinch a narrow part. In three dimensions, one of those fingers would land on top of the object. The failure is therefore better described as grasp reasoning from a single two-dimensional view than as a failure to bind text to coordinates.

5.2. Limitations

The study uses one object-wise split of a small benchmark, RGB input without depth, and no robot trials. Cornell annotations are not exhaustive, so a rectangle that fails the metric could still succeed on a physical object. Development effort was uneven: System B received three architectures, validation-based tuning, and a second training round, whereas Systems C and D each used one frozen prompt. The test split was opened twice for System B, once per round, under rules fixed in advance. System D used one glyph style and one candidate generator. An early formatting check on about 550 grasp rectangles probably included a few images that later entered the test partition, although no parameter or rule was derived from that check. Finally, System A’s correction was prompted by test output, so comparisons that use its 57.7% score are post-hoc and 40.7% remains its clean result. Confirming these findings would require a fresh split or a second dataset.

6. Conclusion

Published accuracies for rule-based, learned, and language-model grasp predictors are difficult to compare because each is usually measured under its own conditions. This paper removed that obstacle for one benchmark. Three systems, one from each paradigm, were given the same images, the same output representation, the same scorer, and a split in which no object is shared between training and test data, and all uncertainty was computed over objects instead of images.

Under those conditions the ranking is clear and statistically robust. A fine-tuned ResNet is correct on roughly four

of five images of unseen objects, a hand-written heuristic on somewhat more than half, and zero-shot GPT-4o on about one in eight. The more useful result is where each system fails. The heuristic finds the object but cannot represent its angle. The network finds the angle and misses on placement. The language model usually fails before either distinction matters, because its rectangle is not near any labeled grasp, and it gives different answers to identical input.

The follow-up experiment changes how that last failure should be understood. Offered a menu of drawn grasps, most of which contained a correct option, GPT-4o chose no better than chance on three different menus and consistently preferred a pinch along the object’s long axis, with high stated confidence. The model’s difficulty is selecting a grasp from a single top-down view, not writing the grasp as numbers. For practitioners, this suggests that wrapping a general-purpose VLM in a better output format will not by itself yield a grasp planner, and that a simple geometric rule on a segmentation mask is a useful baseline: it matches the detector-based heuristic at lower cost and far exceeds the language model.

Two methodological points apply beyond grasping. Clustering the statistics by object widened every interval, and repeating training across seeds removed an architecture effect that a single run had suggested. Both checks cost little and changed what could be claimed. The main open question is whether depth input, or a model with a stronger three-dimensional prior, changes the language model’s choice. All code, the split, the raw model replies, and the scoring scripts are available in the project repository.

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